

# Proposal Proyek Pembelajaran Mesin

S1 Sains Data, Fakultas Matematika Ilmu Pengetahuan dan IPA  
Universitas Negeri Surabaya

|            |                                                                                                                                       |
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| Kelas      | 2024C                                                                                                                                 |
| Ketua      | Kevin Foreman Hutahaeon (24031554095)                                                                                                 |
| Anggota 1  | Kartika Nur Savira (24031554049)                                                                                                      |
| Anggota 2  | Diyanti Pratiwi (24031554016)                                                                                                         |
| URL Github | <a href="https://github.com/Kartika-Nur-Savira/ML_Project9/tree/main">https://github.com/Kartika-Nur-Savira/ML_Project9/tree/main</a> |

## Informasi Umum Proyek

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| Judul                   | Deteksi Dini Kanker Payudara Berdasarkan Fitur Geometri dan Karakteristik Seluler Tumor Menggunakan Ensemble Machine Learning |
| Topik                   | Big Data Kesehatan                                                                                                            |
| Pilihan Rumusan Masalah | 2.6 Pemanfaatan AI dan Big Data Kesehatan                                                                                     |
| Revisi?                 | Ya / Tidak                                                                                                                    |

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| Deskripsi Rumusan masalah      | <p>Kanker payudara merupakan salah satu jenis kanker paling umum pada perempuan dan menjadi penyebab kematian yang signifikan di dunia. Menurut <i>World Health Organization (WHO)</i>, terdapat sekitar 2,3 juta kasus baru kanker payudara dan sekitar 670.000 kematian secara global pada tahun 2022. Tingginya jumlah kasus tersebut menyebabkan deteksi dini menjadi sangat penting untuk meningkatkan peluang kesembuhan pasien. Namun, proses diagnosis kanker payudara masih membutuhkan analisis medis yang kompleks dan waktu yang cukup lama. Oleh karena itu, pemanfaatan <i>Artificial Intelligence (AI)</i> dan <i>Machine Learning</i> dapat menjadi solusi untuk membantu proses klasifikasi kanker payudara secara lebih cepat dan akurat berdasarkan karakteristik tumor dan sel kanker (WHO, 2022).</p> <p>Meskipun demikian, implementasi <i>machine learning</i> pada klasifikasi kanker payudara masih memiliki tantangan karena data medis terdiri dari banyak fitur geometris, tekstur, dan karakteristik biologis yang kompleks. Selain itu, kesalahan klasifikasi terutama <i>false negative</i> pada tumor ganas dapat menyebabkan keterlambatan penanganan pasien dan meningkatkan risiko kematian. Oleh sebab itu, diperlukan model <i>machine learning</i> dengan performa tinggi dan stabil agar dapat membantu proses deteksi dini kanker payudara secara optimal. Penelitian ini menggunakan pendekatan ensemble machine learning untuk melakukan klasifikasi kanker payudara berdasarkan fitur geometri tumor dan karakteristik biologis seluler guna memperoleh model terbaik dalam mendukung implementasi AI pada bidang kesehatan.</p> |
| Tujuan penelitian (Objectives) | <p>Tuliskan tahapan-tahapan, atau target penyelesaian</p> <ol style="list-style-type: none"><li>1. Melakukan proses data understanding dan exploratory data analysis (EDA) pada dataset kanker payudara untuk memahami karakteristik fitur geometri tumor dan karakteristik biologis seluler.</li><li>2. Melakukan tahap data preprocessing meliputi pengecekan missing value, duplikasi data, encoding label, feature scaling, dan pembagian data latih serta data uji untuk mempersiapkan data sebelum proses pemodelan.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|                      | <ol style="list-style-type: none"> <li>3. Membangun model klasifikasi kanker payudara menggunakan beberapa algoritma <i>Machine Learning</i> seperti <i>Logistic Regression</i>, <i>Random Forest</i>, dan <i>XGBoost</i>.</li> <li>4. Membandingkan performa masing-masing algoritma menggunakan metrik evaluasi seperti Accuracy, Precision, Recall, F1-Score, dan Confusion Matrix untuk memperoleh model terbaik.</li> <li>5. Menganalisis fitur-fitur yang paling berpengaruh terhadap klasifikasi kanker payudara berdasarkan hasil evaluasi dan <i>feature importance</i> dari model terbaik.</li> <li>6. Menghasilkan model klasifikasi kanker payudara berbasis <i>ensemble machine learning</i> yang dapat mendukung implementasi <i>Artificial Intelligence (AI)</i> pada bidang kesehatan, khususnya untuk membantu deteksi dini kanker payudara.</li> </ol> |
| Referensi penelitian | <ol style="list-style-type: none"> <li>1. <a href="https://doi.org/10.1109/ICCCNT49239.2020.9225451">https://doi.org/10.1109/ICCCNT49239.2020.9225451</a></li> <li>2. <a href="https://doi.org/10.1016/j.bspc.2021.103279">https://doi.org/10.1016/j.bspc.2021.103279</a></li> <li>3. <a href="https://doi.org/10.1016/j.compbiomed.2021.104931">https://doi.org/10.1016/j.compbiomed.2021.104931</a></li> <li>4. <a href="https://doi.org/10.3233/JIFS-219176">https://doi.org/10.3233/JIFS-219176</a></li> <li>5. <a href="https://doi.org/10.1016/j.irbm.2020.05.005">https://doi.org/10.1016/j.irbm.2020.05.005</a></li> </ol>                                                                                                                                                                                                                                       |

### Informasi Detail Dataset

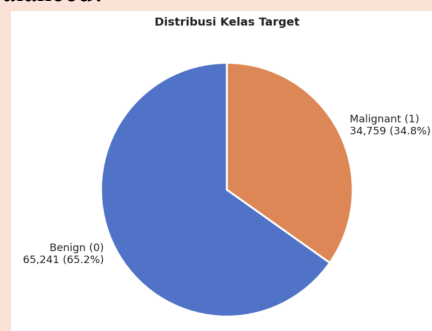
| Sumber dataset               | <a href="https://www.kaggle.com/datasets/sharmajicoder/breast-cancer-dataset">https://www.kaggle.com/datasets/sharmajicoder/breast-cancer-dataset</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                               |                               |                                     |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
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| Deskripsi awal fitur dataset | Dataset yang digunakan terdiri dari 500.000 sampel dengan 40 fitur prediktor dan satu variabel target yaitu diagnosis. Variabel target memiliki dua kelas, yaitu <i>Benign</i> dan <i>Malignant</i> , sehingga termasuk dalam tugas <i>binary classification</i> . Dataset disimpan dalam format CSV dan digunakan untuk membangun model klasifikasi guna memprediksi diagnosis kanker berdasarkan fitur-fitur yang tersedia.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                               |                               |                                     |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | <table><tr><th colspan="4">Tabel 1. Variabel Penelitian</th></tr><tr><th>No</th><th>Variable Name</th><th>Data Type</th><th>Description</th></tr><tr><td>1</td><td><i>radius_mean</i></td><td>Float</td><td>Mean radius of the tumor</td></tr><tr><td>2</td><td><i>texture_mean</i></td><td>Float</td><td>Average texture variation</td></tr><tr><td>3</td><td><i>perimeter_mean</i></td><td>Float</td><td>Mean perimeter of tumor cells</td></tr><tr><td>4</td><td><i>area_mean</i></td><td>Float</td><td>Mean tumor area</td></tr><tr><td>5</td><td><i>smoothness_mean</i></td><td>Float</td><td>Tumor boundary smoothness</td></tr><tr><td>6</td><td><i>compactness_mean</i></td><td>Float</td><td>Compactness of tumor shape</td></tr><tr><td>7</td><td><i>concavity_mean</i></td><td>Float</td><td>Degree of concave portions</td></tr><tr><td>8</td><td><i>concave_points_mean</i></td><td>Float</td><td>Number of concave points</td></tr><tr><td>9</td><td><i>symmetry_mean</i></td><td>Float</td><td>Symmetry of tumor structure</td></tr><tr><td>10</td><td><i>fractal_dimension_mean</i></td><td>Float</td><td>Complexity of tumor boundary</td></tr><tr><td>11</td><td><i>radius_se</i></td><td>Float</td><td>Standard error of radius</td></tr><tr><td>12</td><td><i>texture_se</i></td><td>Float</td><td>Standard error of texture</td></tr><tr><td>13</td><td><i>perimeter_se</i></td><td>Float</td><td>Standard error of perimeter</td></tr><tr><td>14</td><td><i>area_se</i></td><td>Float</td><td>Standard error of area</td></tr><tr><td>15</td><td><i>smoothness_se</i></td><td>Float</td><td>Standard error of smoothness</td></tr><tr><td>16</td><td><i>compactness_se</i></td><td>Float</td><td>Standard error of compactness</td></tr><tr><td>17</td><td><i>concavity_se</i></td><td>Float</td><td>Standard error of concavity</td></tr><tr><td>18</td><td><i>concave_points_se</i></td><td>Float</td><td>Standard error of concave points</td></tr><tr><td>19</td><td><i>symmetry_se</i></td><td>Float</td><td>Standard error of symmetry</td></tr><tr><td>20</td><td><i>fractal_dimension_se</i></td><td>Float</td><td>Standard error of fractal dimension</td></tr><tr><td>21</td><td><i>radius_worst</i></td><td>Float</td><td>Largest radius measurement</td></tr><tr><td>22</td><td><i>texture_worst</i></td><td>Float</td><td>Largest texture measurement</td></tr><tr><td>23</td><td><i>perimeter_worst</i></td><td>Float</td><td>Largest perimeter measurement</td></tr></table> | Tabel 1. Variabel Penelitian  |                               |                                     |  | No | Variable Name | Data Type | Description | 1 | <i>radius_mean</i> | Float | Mean radius of the tumor | 2 | <i>texture_mean</i> | Float | Average texture variation | 3 | <i>perimeter_mean</i> | Float | Mean perimeter of tumor cells | 4 | <i>area_mean</i> | Float | Mean tumor area | 5 | <i>smoothness_mean</i> | Float | Tumor boundary smoothness | 6 | <i>compactness_mean</i> | Float | Compactness of tumor shape | 7 | <i>concavity_mean</i> | Float | Degree of concave portions | 8 | <i>concave_points_mean</i> | Float | Number of concave points | 9 | <i>symmetry_mean</i> | Float | Symmetry of tumor structure | 10 | <i>fractal_dimension_mean</i> | Float | Complexity of tumor boundary | 11 | <i>radius_se</i> | Float | Standard error of radius | 12 | <i>texture_se</i> | Float | Standard error of texture | 13 | <i>perimeter_se</i> | Float | Standard error of perimeter | 14 | <i>area_se</i> | Float | Standard error of area | 15 | <i>smoothness_se</i> | Float | Standard error of smoothness | 16 | <i>compactness_se</i> | Float | Standard error of compactness | 17 | <i>concavity_se</i> | Float | Standard error of concavity | 18 | <i>concave_points_se</i> | Float | Standard error of concave points | 19 | <i>symmetry_se</i> | Float | Standard error of symmetry | 20 | <i>fractal_dimension_se</i> | Float | Standard error of fractal dimension | 21 | <i>radius_worst</i> | Float | Largest radius measurement | 22 | <i>texture_worst</i> | Float | Largest texture measurement | 23 | <i>perimeter_worst</i> | Float | Largest perimeter measurement |
|                              | Tabel 1. Variabel Penelitian                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                               |                               |                                     |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Variable Name                 | Data Type                     | Description                         |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>radius_mean</i>            | Float                         | Mean radius of the tumor            |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>texture_mean</i>           | Float                         | Average texture variation           |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>perimeter_mean</i>         | Float                         | Mean perimeter of tumor cells       |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>area_mean</i>              | Float                         | Mean tumor area                     |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>smoothness_mean</i>        | Float                         | Tumor boundary smoothness           |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>compactness_mean</i>       | Float                         | Compactness of tumor shape          |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>concavity_mean</i>         | Float                         | Degree of concave portions          |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>concave_points_mean</i>    | Float                         | Number of concave points            |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>symmetry_mean</i>          | Float                         | Symmetry of tumor structure         |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>fractal_dimension_mean</i> | Float                         | Complexity of tumor boundary        |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>radius_se</i>              | Float                         | Standard error of radius            |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>texture_se</i>             | Float                         | Standard error of texture           |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>perimeter_se</i>           | Float                         | Standard error of perimeter         |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>area_se</i>                | Float                         | Standard error of area              |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>smoothness_se</i>          | Float                         | Standard error of smoothness        |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>compactness_se</i>         | Float                         | Standard error of compactness       |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>concavity_se</i>           | Float                         | Standard error of concavity         |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>concave_points_se</i>      | Float                         | Standard error of concave points    |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>symmetry_se</i>            | Float                         | Standard error of symmetry          |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>fractal_dimension_se</i>   | Float                         | Standard error of fractal dimension |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
|                              | 21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>radius_worst</i>           | Float                         | Largest radius measurement          |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
| 22                           | <i>texture_worst</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Float                         | Largest texture measurement   |                                     |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |
| 23                           | <i>perimeter_worst</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Float                         | Largest perimeter measurement |                                     |  |    |               |           |             |   |                    |       |                          |   |                     |       |                           |   |                       |       |                               |   |                  |       |                 |   |                        |       |                           |   |                         |       |                            |   |                       |       |                            |   |                            |       |                          |   |                      |       |                             |    |                               |       |                              |    |                  |       |                          |    |                   |       |                           |    |                     |       |                             |    |                |       |                        |    |                      |       |                              |    |                       |       |                               |    |                     |       |                             |    |                          |       |                                  |    |                    |       |                            |    |                             |       |                                     |    |                     |       |                            |    |                      |       |                             |    |                        |       |                               |

|                           | <div><div>24</div><div>area_worst</div><div>Float</div><div>Largest tumor area observed</div></div> <div><div>25</div><div>smoothness_worst</div><div>Float</div><div>Worst smoothness value</div></div> <div><div>26</div><div>compactness_worst</div><div>Float</div><div>Worst compactness value</div></div> <div><div>27</div><div>concavity_worst</div><div>Float</div><div>Worst concavity value</div></div> <div><div>28</div><div>concave_points_worst</div><div>Float</div><div>Worst concave points value</div></div> <div><div>29</div><div>symmetry_worst</div><div>Float</div><div>Worst symmetry measurement</div></div> <div><div>30</div><div>fractal_dimension_worst</div><div>Float</div><div>Worst fractal dimension</div></div> <div><div>31</div><div>cell_density</div><div>Float</div><div>Density of cells in tumor tissue</div></div> <div><div>32</div><div>nucleus_size</div><div>Float</div><div>Average size of cell nuclei</div></div> <div><div>33</div><div>nucleus_texture</div><div>Float</div><div>Variation in nucleus texture</div></div> <div><div>34</div><div>mitosis_rate</div><div>Float</div><div>Rate of cell division</div></div> <div><div>35</div><div>tumor_border_irregularity</div><div>Float</div><div>Irregularity of tumor borders</div></div> <div><div>36</div><div>chromatin_density</div><div>Float</div><div>Chromatin concentration in nucleus</div></div> <div><div>37</div><div>cell_variation</div><div>Float</div><div>Variation in cell sizes</div></div> <div><div>38</div><div>nuclear_variation</div><div>Float</div><div>Variation in nucleus shapes</div></div> <div><div>39</div><div>cytoplasm_ratio</div><div>Float</div><div>Cytoplasm to nucleus ratio</div></div> <div><div>40</div><div>cell_clump_thickness</div><div>Float</div><div>Thickness of cell clusters</div></div> <div><div>41</div><div>diagnosis</div><div>Object /<br/>Categoric</div><div>Tumor diagnosis result (Benign /<br/>Malignant)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |        |          |          |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|----------|----------|-----|----------|-------------|--------|-------|-------|--------|-------|--------------|--------|-------|-------|--------|--------|----------------|--------|--------|-------|---------|-------|-----------|---------|---------|-------|----------|-------|-----------------|--------|-------|--------|-------|--------|------------------|-------|-------|--------|-------|-------|----------------|-------|-------|--------|-------|--------|---------------------|-------|-------|--------|-------|--------|---------------|-------|-------|--------|-------|--------|------------------------|-------|-------|--------|-------|--------|-----------|--------|-------|--------|-------|-------|------------|-------|-------|--------|-------|-------|--------------|--------|-------|--------|-------|-------|---------|-------|-------|--------|-------|--------|---------------|--------|-------|--------|-------|-------|----------------|-------|-------|--------|-------|-------|--------------|-------|-------|--------|-------|--------|-------------------|-------|-------|--------|-------|--------|-------------|-------|-------|--------|-------|--------|
| Aspek khusus dari dataset | <div><div>1. Apakah datanya sparse?</div><div>Berdasarkan hasil pengecekan jumlah missing value pada data untuk semua kolom adalah 0, atau persentase <i>sparsity</i> dalam konteks kelengkapan data adalah <b>0%</b>. Data didominasi oleh tipe data float yang berisi angka desimal.</div></div> <div><div>2. Bagaimana ringkasan statistik deskriptifnya?</div><div><div>Tabel 2. Ringkasan statistik deskriptif</div><table><tr><th>Variabel</th><th>Mean</th><th>Std</th><th>Min</th><th>Max</th><th>Skewness</th></tr><tr><td>radius_mean</td><td>13.998</td><td>3.003</td><td>0.168</td><td>28.689</td><td>0.004</td></tr><tr><td>texture_mean</td><td>19.998</td><td>5.001</td><td>0.024</td><td>43.231</td><td>-0.004</td></tr><tr><td>perimeter_mean</td><td>83.975</td><td>19.362</td><td>0.909</td><td>179.970</td><td>0.156</td></tr><tr><td>area_mean</td><td>643.922</td><td>267.268</td><td>0.088</td><td>2585.669</td><td>0.634</td></tr><tr><td>smoothness_mean</td><td>-0.000</td><td>0.998</td><td>-4.872</td><td>4.814</td><td>-0.001</td></tr><tr><td>compactness_mean</td><td>0.000</td><td>1.001</td><td>-4.574</td><td>5.220</td><td>0.001</td></tr><tr><td>concavity_mean</td><td>0.000</td><td>0.999</td><td>-4.573</td><td>5.022</td><td>-0.003</td></tr><tr><td>concave_points_mean</td><td>0.001</td><td>1.000</td><td>-4.677</td><td>4.655</td><td>-0.005</td></tr><tr><td>symmetry_mean</td><td>0.001</td><td>1.000</td><td>-4.482</td><td>4.611</td><td>-0.000</td></tr><tr><td>fractal_dimension_mean</td><td>0.001</td><td>1.000</td><td>-4.625</td><td>4.575</td><td>-0.000</td></tr><tr><td>radius_se</td><td>-0.000</td><td>1.001</td><td>-4.980</td><td>4.719</td><td>0.001</td></tr><tr><td>texture_se</td><td>0.003</td><td>1.001</td><td>-4.717</td><td>4.798</td><td>0.007</td></tr><tr><td>perimeter_se</td><td>-0.002</td><td>1.000</td><td>-4.829</td><td>5.076</td><td>0.000</td></tr><tr><td>area_se</td><td>0.002</td><td>0.999</td><td>-4.535</td><td>4.732</td><td>-0.004</td></tr><tr><td>smoothness_se</td><td>-0.000</td><td>1.001</td><td>-5.306</td><td>4.920</td><td>0.004</td></tr><tr><td>compactness_se</td><td>0.001</td><td>1.000</td><td>-4.631</td><td>4.738</td><td>0.000</td></tr><tr><td>concavity_se</td><td>0.001</td><td>1.001</td><td>-4.317</td><td>4.786</td><td>-0.004</td></tr><tr><td>concave_points_se</td><td>0.000</td><td>1.000</td><td>-4.581</td><td>4.679</td><td>-0.001</td></tr><tr><td>symmetry_se</td><td>0.002</td><td>1.000</td><td>-4.480</td><td>4.795</td><td>-0.000</td></tr></table></div></div> | Variabel | Mean   | Std      | Min      | Max | Skewness | radius_mean | 13.998 | 3.003 | 0.168 | 28.689 | 0.004 | texture_mean | 19.998 | 5.001 | 0.024 | 43.231 | -0.004 | perimeter_mean | 83.975 | 19.362 | 0.909 | 179.970 | 0.156 | area_mean | 643.922 | 267.268 | 0.088 | 2585.669 | 0.634 | smoothness_mean | -0.000 | 0.998 | -4.872 | 4.814 | -0.001 | compactness_mean | 0.000 | 1.001 | -4.574 | 5.220 | 0.001 | concavity_mean | 0.000 | 0.999 | -4.573 | 5.022 | -0.003 | concave_points_mean | 0.001 | 1.000 | -4.677 | 4.655 | -0.005 | symmetry_mean | 0.001 | 1.000 | -4.482 | 4.611 | -0.000 | fractal_dimension_mean | 0.001 | 1.000 | -4.625 | 4.575 | -0.000 | radius_se | -0.000 | 1.001 | -4.980 | 4.719 | 0.001 | texture_se | 0.003 | 1.001 | -4.717 | 4.798 | 0.007 | perimeter_se | -0.002 | 1.000 | -4.829 | 5.076 | 0.000 | area_se | 0.002 | 0.999 | -4.535 | 4.732 | -0.004 | smoothness_se | -0.000 | 1.001 | -5.306 | 4.920 | 0.004 | compactness_se | 0.001 | 1.000 | -4.631 | 4.738 | 0.000 | concavity_se | 0.001 | 1.001 | -4.317 | 4.786 | -0.004 | concave_points_se | 0.000 | 1.000 | -4.581 | 4.679 | -0.001 | symmetry_se | 0.002 | 1.000 | -4.480 | 4.795 | -0.000 |
| Variabel                  | Mean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Std      | Min    | Max      | Skewness |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| radius_mean               | 13.998                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3.003    | 0.168  | 28.689   | 0.004    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| texture_mean              | 19.998                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5.001    | 0.024  | 43.231   | -0.004   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| perimeter_mean            | 83.975                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 19.362   | 0.909  | 179.970  | 0.156    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| area_mean                 | 643.922                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 267.268  | 0.088  | 2585.669 | 0.634    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| smoothness_mean           | -0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.998    | -4.872 | 4.814    | -0.001   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| compactness_mean          | 0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.001    | -4.574 | 5.220    | 0.001    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| concavity_mean            | 0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.999    | -4.573 | 5.022    | -0.003   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| concave_points_mean       | 0.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.000    | -4.677 | 4.655    | -0.005   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| symmetry_mean             | 0.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.000    | -4.482 | 4.611    | -0.000   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| fractal_dimension_mean    | 0.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.000    | -4.625 | 4.575    | -0.000   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| radius_se                 | -0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1.001    | -4.980 | 4.719    | 0.001    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| texture_se                | 0.003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.001    | -4.717 | 4.798    | 0.007    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| perimeter_se              | -0.002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1.000    | -4.829 | 5.076    | 0.000    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| area_se                   | 0.002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.999    | -4.535 | 4.732    | -0.004   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| smoothness_se             | -0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1.001    | -5.306 | 4.920    | 0.004    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| compactness_se            | 0.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.000    | -4.631 | 4.738    | 0.000    |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| concavity_se              | 0.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.001    | -4.317 | 4.786    | -0.004   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| concave_points_se         | 0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.000    | -4.581 | 4.679    | -0.001   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |
| symmetry_se               | 0.002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.000    | -4.480 | 4.795    | -0.000   |     |          |             |        |       |       |        |       |              |        |       |       |        |        |                |        |        |       |         |       |           |         |         |       |          |       |                 |        |       |        |       |        |                  |       |       |        |       |       |                |       |       |        |       |        |                     |       |       |        |       |        |               |       |       |        |       |        |                        |       |       |        |       |        |           |        |       |        |       |       |            |       |       |        |       |       |              |        |       |        |       |       |         |       |       |        |       |        |               |        |       |        |       |       |                |       |       |        |       |       |              |       |       |        |       |        |                   |       |       |        |       |        |             |       |       |        |       |        |

|                           |         |         |        |          |        |
|---------------------------|---------|---------|--------|----------|--------|
| fractal_dimension_se      | -0.001  | 1.000   | -4.513 | 4.739    | -0.003 |
| radius_worst              | 16.015  | 3.154   | 0.953  | 31.157   | 0.010  |
| texture_worst             | 0.001   | 0.999   | -4.672 | 4.984    | -0.005 |
| perimeter_worst           | -0.003  | 0.999   | -4.628 | 4.483    | -0.001 |
| area_worst                | 837.231 | 374.642 | 0.130  | 4010.403 | 0.821  |
| smoothness_worst          | 0.001   | 0.999   | -4.534 | 5.129    | -0.001 |
| compactness_worst         | 0.000   | 0.999   | -4.394 | 4.577    | 0.000  |
| concavity_worst           | 0.000   | 1.000   | -4.622 | 5.034    | -0.003 |
| concave_points_worst      | -0.001  | 1.000   | -4.824 | 4.671    | -0.001 |
| symmetry_worst            | -0.000  | 1.001   | -4.812 | 4.358    | -0.005 |
| fractal_dimension_worst   | -0.001  | 1.000   | -4.381 | 4.315    | -0.004 |
| cell_density              | 49.984  | 14.987  | 0.001  | 122.558  | 0.005  |
| nucleus_size              | 10.001  | 2.993   | 0.003  | 23.976   | 0.010  |
| nucleus_texture           | -0.001  | 1.002   | -4.809 | 5.020    | 0.001  |
| mitosis_rate              | 4.034   | 1.931   | 0.000  | 13.583   | 0.187  |
| tumor_border_irregularity | 0.500   | 0.198   | 0.000  | 1.398    | 0.072  |
| chromatin_density         | -0.002  | 0.999   | -4.639 | 4.580    | -0.003 |
| cell_variation            | 0.000   | 1.000   | -4.378 | 4.434    | -0.003 |
| nuclear_variation         | 0.001   | 1.000   | -4.632 | 4.828    | 0.005  |
| cytoplasm_ratio           | 0.001   | 0.999   | -5.195 | 4.679    | -0.002 |
| cell_clump_thickness      | -0.001  | 1.001   | -5.119 | 4.913    | 0.003  |

Berdasarkan statistik deskriptif pada Tabel 2, sebagian besar variabel memiliki mean mendekati 0 dan standar deviasi sekitar 1 karena telah distandarisasi. Nilai skewness mayoritas juga mendekati 0 sehingga distribusi data cenderung simetris atau mendekati normal. Namun, beberapa variabel seperti 'are\_mean' dan 'area\_worst' menunjukkan skewness positif.

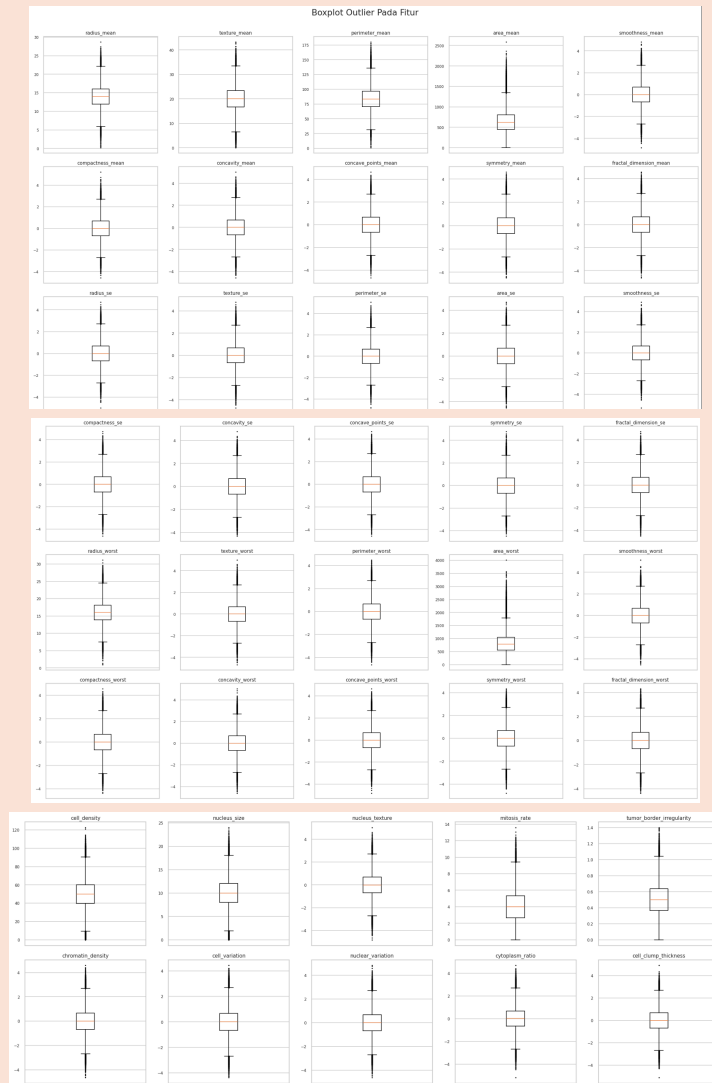
3. Apakah datanya imbalanced?



**Gambar 1.** Distribusi Kelas Target

Distribusi kelas target pada variabel diagnosis menunjukkan bahwa data didominasi oleh kelas benign (0) dengan sebaran sebanyak 62.241 (65.2%), sedangkan pada kelas malignant (1) jumlah data lebih sedikit sebanyak 34.759 (34.8%), dengan rasio antara keduanya yaitu 2:1. Kondisi ini menunjukkan adanya *class imbalance* ringan, karena jumlah data pada kelas benign lebih banyak dibandingkan kelas malignant.

4. Bagaimana outlier nya?



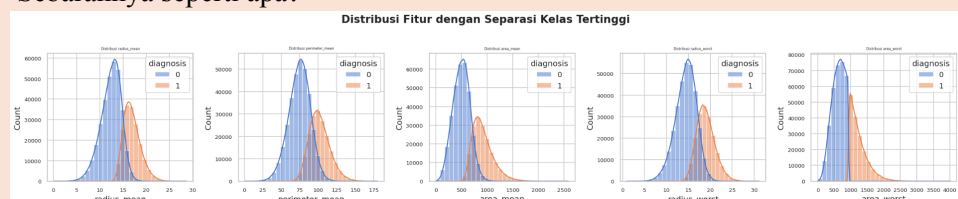
**Gambar 2.** Distribusi Outlier Pada Fitur

Pada visualisasi *boxplot* terlihat adanya *outlier* pada setiap variabel yang ditandai dengan adanya titik diluar *whisker*, *outlier* paling banyak muncul pada variabel *area\_mean*, *area\_worst*, *radius\_mean*, dan *radius\_worst*, terutama pada kelas *Malignant*, yang menunjukkan adanya tumor dengan ukuran jauh lebih besar dibanding data lainnya. Sementara itu, variabel lainnya memiliki penyebaran *outlier* yang relatif seimbang pada kedua kelas. *Outlier* pada fitur-fitur ini memiliki rentang 1700 hingga 8800 data outlier, namun *outlier* masih wajar karena perbedaan kondisi pada tiap pasien.

5. Apakah ada data yang terduplikat?

Berdasarkan hasil pengecekan duplikasi pada data untuk semua kolom adalah 0, artinya tidak terdapat baris data yang identik secara keseluruhan dalam dataset. Setiap sampel merupakan data yang unik sehingga tidak diperlukan proses penghapusan duplikat.

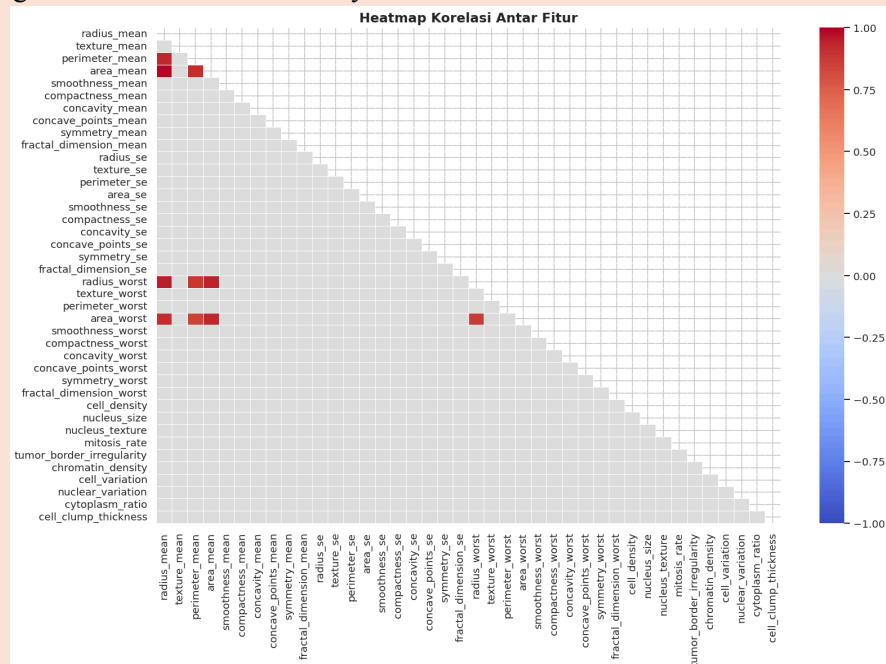
6. Sebarannya seperti apa?



### Gambar 3. Distribusi Fitur dengan Separasi Kelas Tertinggi

Dari hasil eksplorasi distribusi seluruh fitur terlihat bahwa semua fitur berdistribusi normal, visualisasi berikut menampilkan lima fitur dengan separasi kelas tertinggi antara benign dan malignant, yaitu 'radius\_mean', 'perimeter\_mean', 'area\_mean', 'radius\_worst', dan area\_worst. Berdasarkan visualisasi, kelas benign (0) cenderung memiliki nilai yang lebih rendah dan terpusat, sedangkan kelas malignant (1) memiliki nilai lebih tinggi dengan sebaran yang lebih lebar. Hal ini mengindikasikan bahwa tumor ganas secara umum memiliki ukuran radius, perimeter, dan area yang lebih besar dibandingkan tumor jinak.

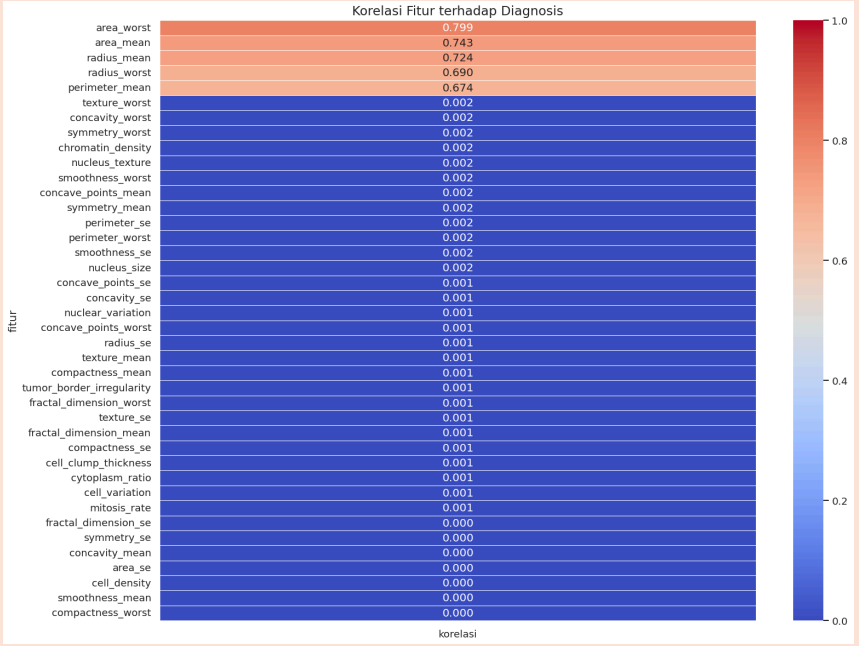
#### 7. Bagaimana korelasi antar fiturnya?



### Gambar 4. Korelasi Antar Fitur

Berdasarkan heatmap korelasi antar fitur pada Gambar 4, terlihat bahwa sebagian besar fitur memiliki korelasi rendah satu sama lain (ditunjukkan dengan warna abu-abu). Namun terdapat beberapa pasang fitur dengan korelasi sangat tinggi (warna merah gelap), yaitu antara 'radius\_mean', 'perimeter\_mean', dan 'area\_mean', serta antara 'radius\_worst', 'perimeter\_worst', dan 'area\_worst'. Hal ini wajar karena ketiga fitur tersebut memang saling berkaitan semakin besar radius, maka perimeter dan area tumor juga akan ikut membesar.

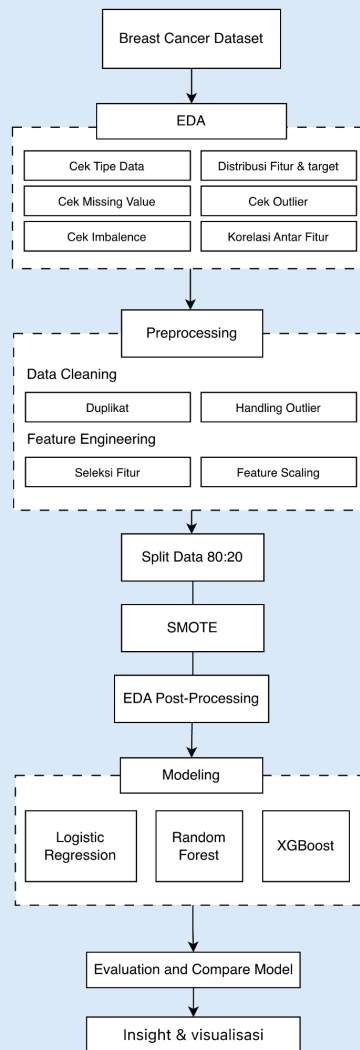
#### 8. Bagaimana korelasi fitur dengan target?

|  |                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  <p><b>Gambar 5. Korelasi Fitur Dengan Target</b></p> <p>Berdasarkan hasil analisis pada Gambar 5, hanya lima fitur yang menunjukkan korelasi signifikan yaitu ‘area_worst’, ‘area_mean’, ‘radius_mean’, ‘radius_worst’, ‘perimeter_worst’, dan ‘perimeter_mean’. Kelima fitur ini merupakan fitur berbasis ukuran geometri tumor.</p> |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Rencana Metode

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teknik yang digunakan             | <i>Logistic Regression (baseline)</i><br><i>Random Forest (ensemble bagging)</i><br><i>XGBoost / Gradient Boosting (ensemble boosting)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Alasan pemilihan teknik/algoritma | <p><b>Logistic Regression</b> dipilih sebagai model baseline karena memiliki kompleksitas komputasi rendah sehingga efisien dijalankan pada dataset berskala besar dengan 500.000 observasi, model bersifat linear dan mudah untuk diinterpretasikan sehingga dapat dijadikan acuan minimum performa yang harus dilampaui oleh model ensemble.</p> <p><b>Random Forest</b> dipilih karena merupakan metode ensemble berbasis bagging yang mampu menangani data tabular berukuran besar, dan random forest sendiri terbukti robust terhadap outlier sehingga mengurangi risiko overfitting.</p> <p><b>XGBoost</b> dipilih karena merupakan metode boosting yang memiliki performa tinggi dalam klasifikasi, mampu mempelajari pola kompleks seperti pada dataset kami fitur geometri, tekstur dan karakteristik biologis sel yang saling berkaitan sehingga <i>XGBoost</i> digunakan untuk melihat pola datanya, <i>XGBoost</i> juga digunakan untuk memperbaiki kesalahan prediksi menggunakan <i>boosting</i> sehingga nanti memiliki performa yang stabil, serta sering memberikan hasil akurasi terbaik pada data medis dan tidak seimbang.</p> |
| Langkah-langkah umum pengerjaan   | <p>Tuliskan gambaran awal Anda menyelesaikan proses latih model tersebut.</p> <p>Metrik apa yang Anda gunakan?</p> <p><b>1. Diagram Alur</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |





1. Memahami dan mengeksplorasi dataset melakukan pengecekan struktur data, missing value, cek outlier, cek imbalance, dan distribusi fitur.
2. Melakukan Preprocessing data cleaning (Handling outlier), serta melakukan feature *engineering* seleksi fitur dan scaling untuk menyamakan bobot data yang mungkin berbeda.
3. Membagi dataset menjadi data latih (80) dan data uji (20).
4. Melakukan EDA post processing untuk melihat distribusi data setelah dilakukan *preprocessing*.
5. Melatih tiga model: Logistic Regression, Random Forest, dan XGBoost / Gradient Boosting.
6. Melakukan evaluasi performa masing-masing model pada data uji.
7. Membandingkan hasil performa ketiga model.  
Metrik Evaluasi yang Digunakan  
Accuracy  
Precision  
Recall  
F1-score  
Confusion Matrix
8. Membuat insight dan visualisasi